

Performing a TCP and UDP connection for **client – server** communication using terminal (NETCAT)

Writeup

Netcat is a profound tool used by network analysts to read and write data across network connections, using TCP / UPD transport protocols. Designed as a reliable “back-end” tool, Netcat can be used directly with other programs and scripts to send files from a client to a server and back. At the same time, it’s a feature-rich network debugging and exploration tool that can specify network parameters while also establishing a connection to a remote host via a tunnel.

Netcat can perform multiple functions but its main role is to –

- Create an initial socket to establish connection
- Once connected, netcat will automatically generate a second socket to transmit files

Netcat is mainly used in Linux (nc) but since we are using windows here, the keyword here is ncat.

Syntax:

```
PS C:\Users\APPADI> ncat -h
Ncat 7.99 ( https://nmap.org/ncat )
Usage: ncat [options] [hostname] [port]
```

Basic parameters:

- **-l:** set the “listen” mode, waits for the incoming connections.
- **-p:** local port.
- **-u:** set the UDP mode.
- **-v:** Verbose mode (shows connection details)
- **-p:** Specify the local source port

FOR TCP

Step 1:

Open two terminals in your pc, the first will act as your server and the second will be you client.

Step 2:

With Netcat, the PC can be converted into a server by listening on port 2399 :

```
PS C:\Users\APPADI> ncat -l 2399
|
```

Step 3:

Alongside on the other terminal you can connect to the recently opened port 2399 as the client:

```
PS C:\Users\APPADI> ncat localhost 2399
|
```

Step 4:

Since the connection has been established now we can write data from the client side and the text will appear in you server seamlessly.

```
PS C:\Users\APPADI> ncat -l 2399
Hello!!
I am Rohit Appadi
Currently testing TCP connection using netcat
|
```

Server Side

```
PS C:\Users\APPADI> ncat localhost 2399
Hello!!
I am Rohit Appadi
Currently testing TCP connection using netcat
|
```

Client Side

FOR UDP

Step 1:

On the **server** side, run the command below. As you can see, you only need to add **-u** to the command to establish the **UDP** connection:

```
PS C:\Users\APPADI> ncat -u -l 2399
|
```

Step 2:

On the **client** side, run the command below. As you can see, you only need to add **-u** to the command to establish the **UDP** connection

```
PS C:\Users\APPADI> ncat -u localhost 2399
|
```

Step 3:

On the client side, write the text and it will appear on the server side seamlessly.

Step 4:

To check for UDP connection between client and server, we can verify on a 3rd terminal using netstat:

```
PS C:\Users\APPADI> netstat -an | Select-String 2399

UDP      [::]:2399          [::1]:58256
UDP      [::]:58256         [::1]:2399
```

And as we can see below, the text appeared seamlessly on both ends:

```
PS C:\Users\APPADI> ncat -u -l 2399
Hello!!
I am Rohit Appadi
Testing UDP connection
|
```

Server Side

```
PS C:\Users\APPADI> ncat -u localhost 2399
Hello!!
I am Rohit Appadi
Testing UDP connection
```

Client Side